



PHẠM PHƯỚC TRƯỜNG

FLUTTER DEVELOPER

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SUMMARY

Flutter Developer with hands-on production experience building complex video streaming systems for mobile. Specialized in high-performance mobile architecture — including HLS local proxy servers, Adaptive Bitrate (ABR) pipelines, and multi-layer caching strategies. Strong foundation in async/concurrent programming, race condition prevention, and memory optimization for long video sessions. Consistently prioritizes Clean Architecture, SOLID principles, and long-term maintainability across every feature built.

WORK EXPERIENCE

Mobile Developer - CSS Can Tho

May 2024 - Present

- Architected a production HLS system with a local proxy server, buffer-aware grace timer, and 4-phase offline recovery — ensuring uninterrupted playback on unstable networks.
- Built a video player object pool with epoch-based concurrency, debounced prefetch, and platform-adaptive sizing (iOS/Android) — delivering frame-ready playback on every swipe.
- Decomposed a large multi-concern controller into focused, independently testable units via mixin composition, each owning a single responsibility (state, lifecycle, events, network).
- Implemented a custom async semaphore + generation-guard pattern to eliminate race conditions during rapid swiping, preventing native-layer crashes from stale init tasks.
- Built an offline-first progress tracker with local persistence, batch sync (≤10/session), exponential-backoff retry, and atomic read/write — zero data loss on sudden app termination.
- Engineered a multi-layer RAM cache with explicit eviction and bounded collections to reduce GC pressure and prevent OOM errors in long video sessions.
- Designed a navigation-stack-based playback coordinator that propagates interaction state (like, save, comment, view count) across all active screens via snapshot-safe iteration.
- Integrated an ABR engine with EWMA bandwidth estimation and quality capping during preload — preventing bandwidth contention between active playback and background prefetch.
- Applied Clean Architecture (data / domain / presentation) with interface-driven design and GetX-based dependency injection across all feature modules.

SKILLS

Mobile	Flutter (Dart) · Android Native (Platform Channels)
Architecture	Clean Architecture · SOLID · Mixin-based Decomposition
State Mgmt	GetX (Reactive + DI) · BLoC
Async / Concurrency	Custom Semaphore · Generation Guard · CancelableOperation
Video Streaming	HLS (m3u8/ts) · Local Proxy · ABR · VideoPlayerController
Networking	RESTful APIs · Dio · HTTP Socket · Retry / Backoff
Performance	Object Pool · RAM Eviction · CPU/GPU Profiling · Thermal Control
Storage	SharedPreferences · SQLite · File I/O · Firebase Realtime DB
Delivery	Git · Google Play Console · CI/CD basics
Languages	English

FEATURED PROJECTS

Click-ed - Educational Social Network

[Google Play](#)

A TikTok-style short-video learning platform with HLS adaptive streaming, offline progress tracking, and a continuous-scroll feed for educational content.

- **Video Architecture:** Epoch-based object pool + debounced prefetch + warm-up pre-init — video is frame-ready the moment the user finishes swiping.
- **Network Resilience:** 4-phase offline recovery with buffer-aware grace timer (3s–60s), early buffer-drain detection, and auto-resume at the correct playback position on reconnect.
- **Local Proxy & Cache:** HLS proxy server with multi-layer cache (in-memory + disk), lazy playlist rebuild from downloaded segments, and bounded disk eviction.
- **Adaptive Bitrate:** EWMA bandwidth estimation + network-type safety factors + quality capping during preload — eliminates quality thrashing under fluctuating connections.
- **Offline Tracking:** Batch sync, exponential-backoff retry, atomic read/write, and checkpoint fallback to handle edge cases where the player resets position to zero while offline.
- **Global State:** Navigation-stack coordinator propagates like/save/comment/view count updates to all active screens simultaneously via snapshot-safe iteration.

MyDTU - Student Portal · Duy Tan University

[Google Play](#)

A university app for checking schedules, grades, and campus services, serving thousands of active students.

- Restructured legacy codebase to Clean Architecture, improving scalability without introducing regression.
- Built local caching + request deduplication, reducing server load and improving cold-start response time.
- Perform ongoing maintenance and occasional feature development to ensure app stability, resolve bugs, and adapt to new university requirements.
- Managed full release cycle on Google Play Console: versioning, policy compliance, staged rollout.

Linkeep - Link Management & Social Sharing

[Google Play](#)

A cross-platform Flutter app for saving and sharing links with offline-first architecture, secure authentication, and real-time sync.

- SQLite + in-memory cache with write-through sync to Firebase Realtime DB — consistent cross-device data, zero perceptible lag.
- Deep link / QR / email sharing with accurate screen routing and reliable external browser fallback.
- PIN/biometric security with per-category private access control.
- WebView cache-first loading, ad/popup blocking, automatic URL parsing from clipboard — instant link intake.

EDUCATION

Can Tho University - Bachelor of Engineering in Information Systems

Relevant coursework: Object-Oriented Programming, Software Architecture, System Analysis and Design, Database Management.